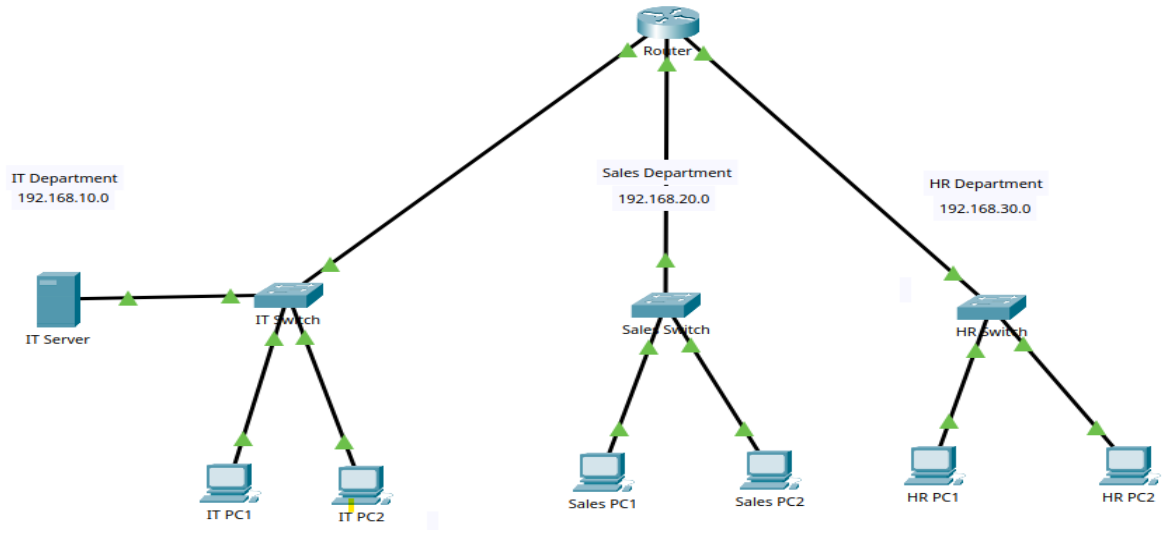
Name: Sunday Tanimowo

Student ID: CS-C26-09-NA-006

Date: Tuesday 6th October 2026

TechNova Ltd. Network
Using Cisco Packet Tracer To Build

The Topology:

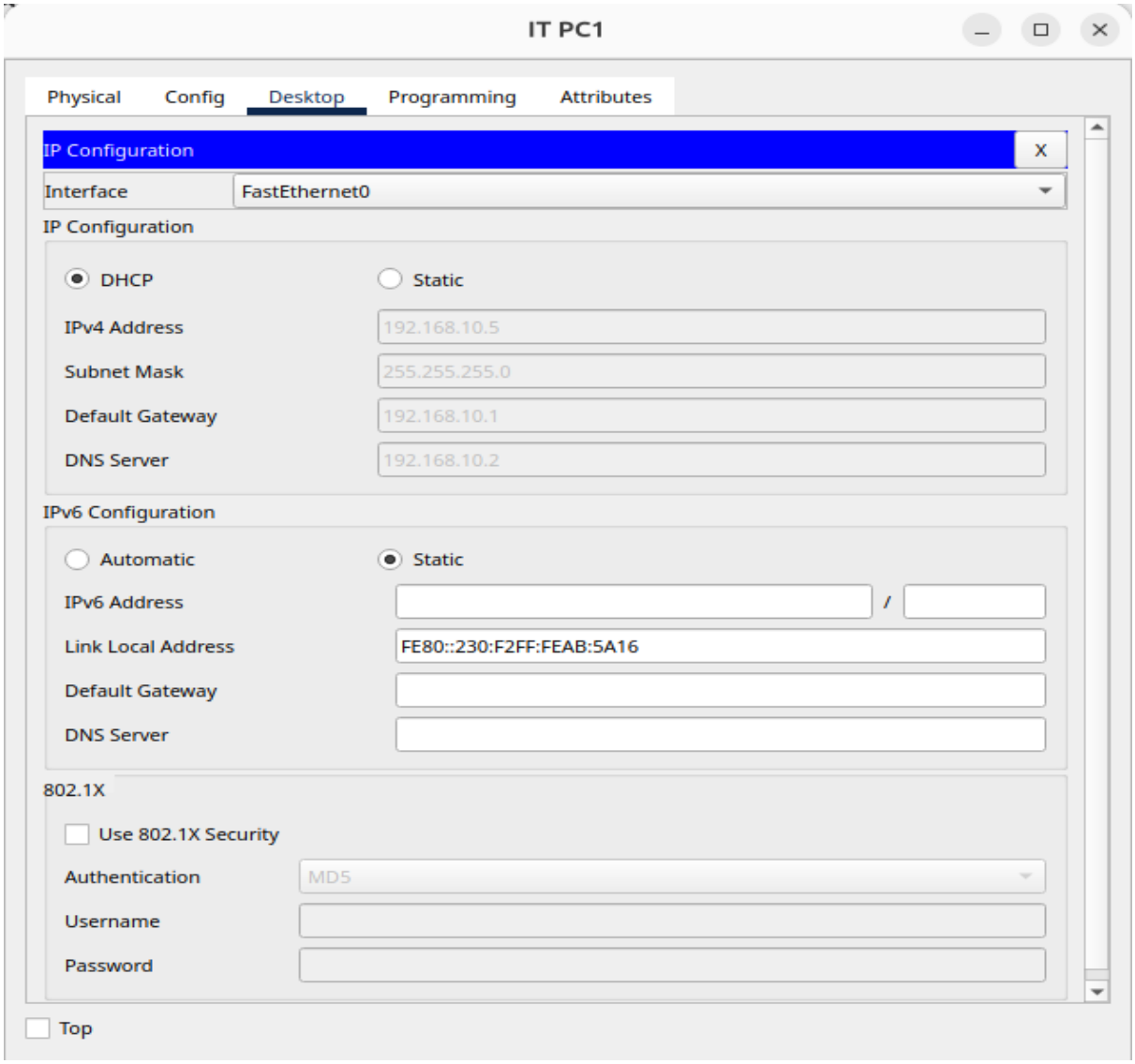


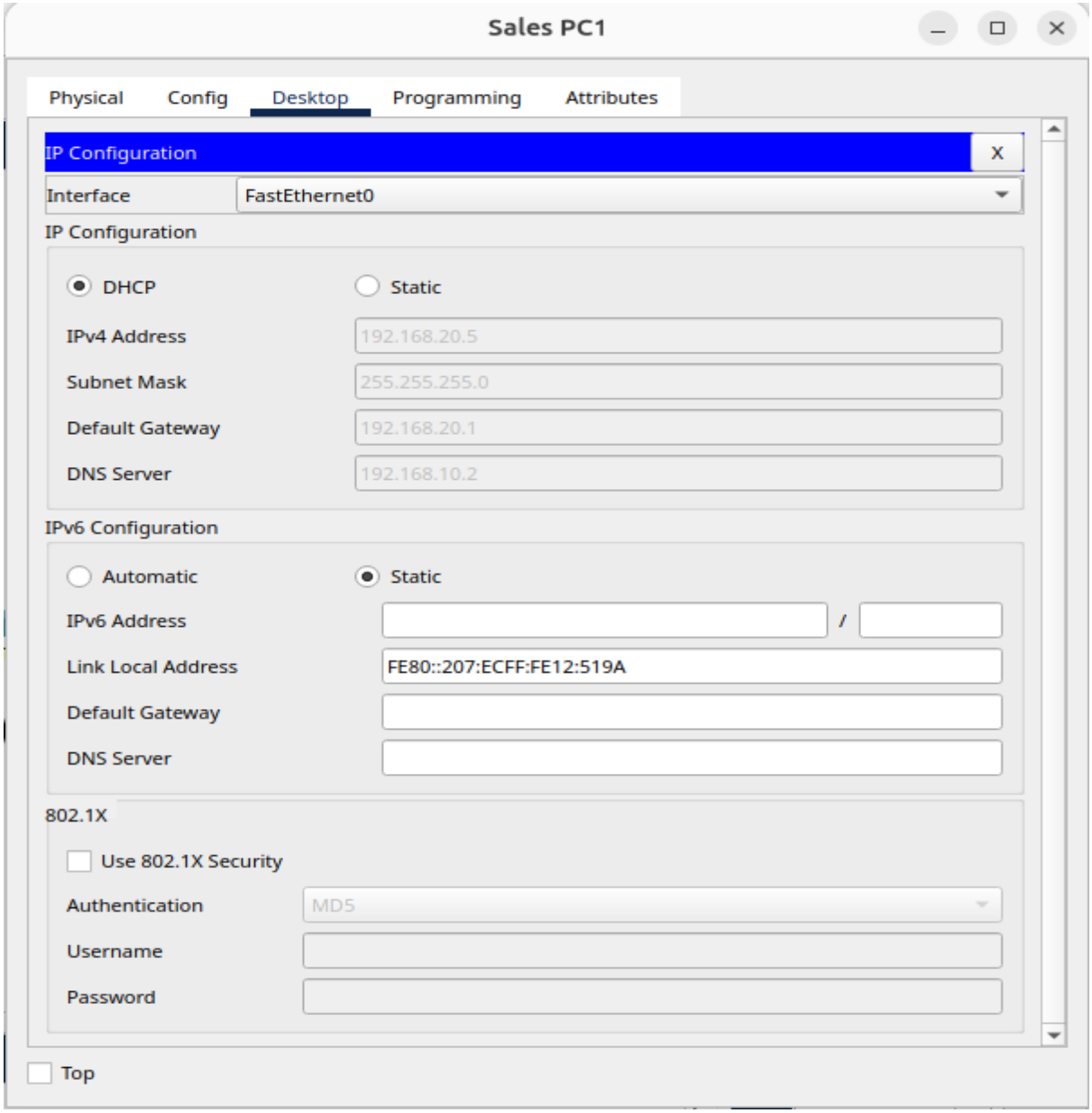
The topology above represents the network infrastructure of **TechNova Limited**, comprising three departments: **IT, Sales, and HR**.

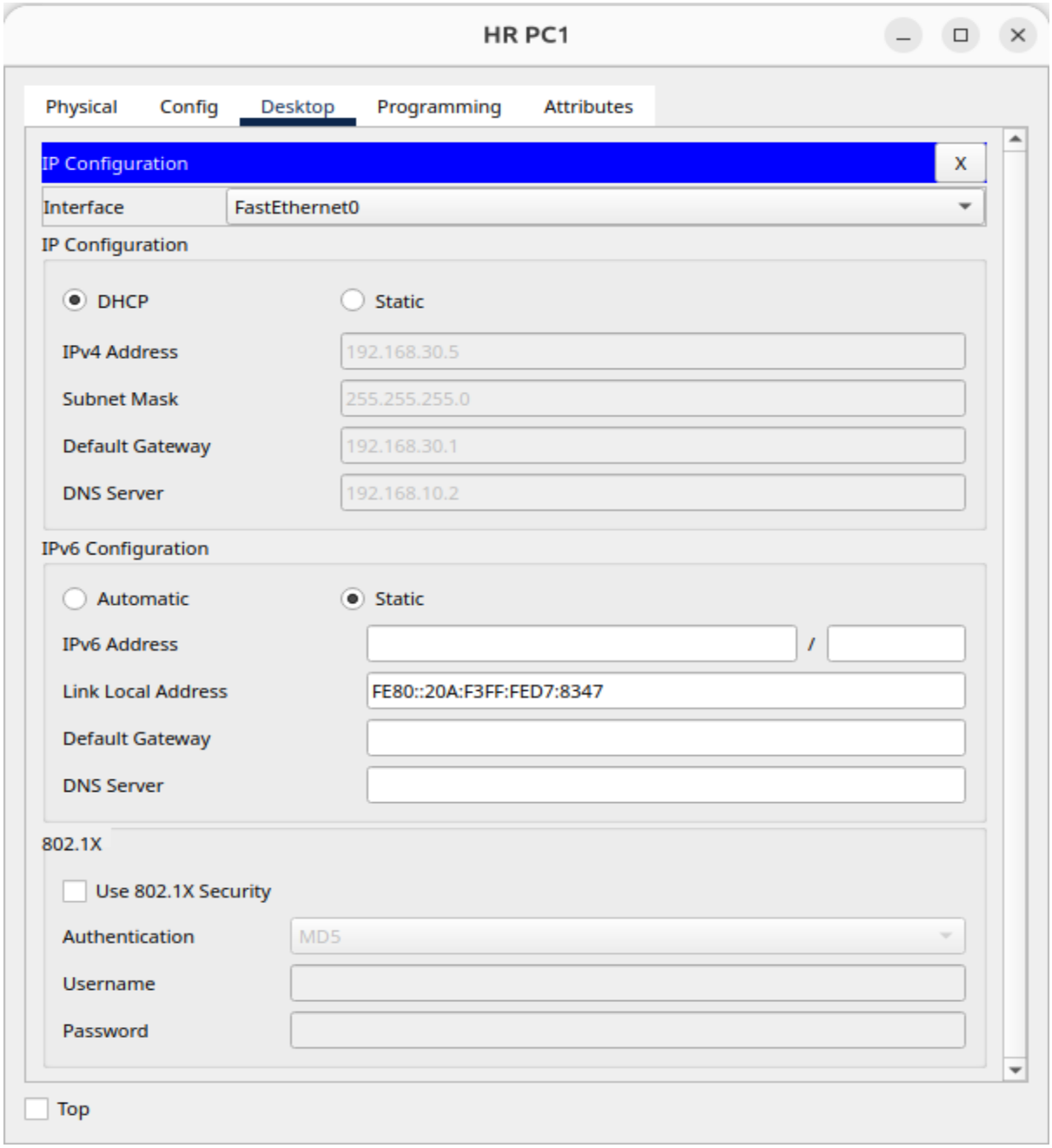
The **IT department** consists of two PCs, a switch, and a server, while the **Sales** and **HR departments** each consist of two PCs and a switch. A **router** is used to connect the three departmental networks and facilitate communication between them.

The IT department operates on the **192.168.10.0/24** network, the Sales department operates on the **192.168.20.0/24** network, and the HR department operates on the **192.168.30.0/24** network. The **IT server is configured as the DHCP server**, providing automatic IP addressing and other network configuration information to the PCs across the three departmental networks.

IP Address Configurations:

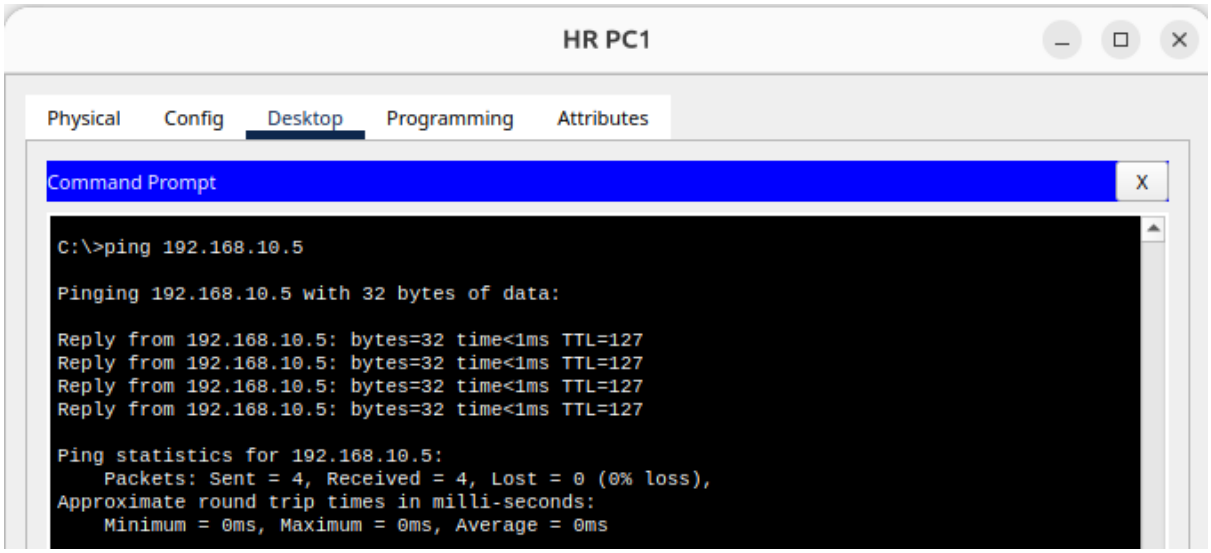



Sales PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.20.5
Subnet Mask
255.255.255.0
Default Gateway
192.168.20.1
DNS Server
192.168.10.2
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::207:ECFF:FE12:519A
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top



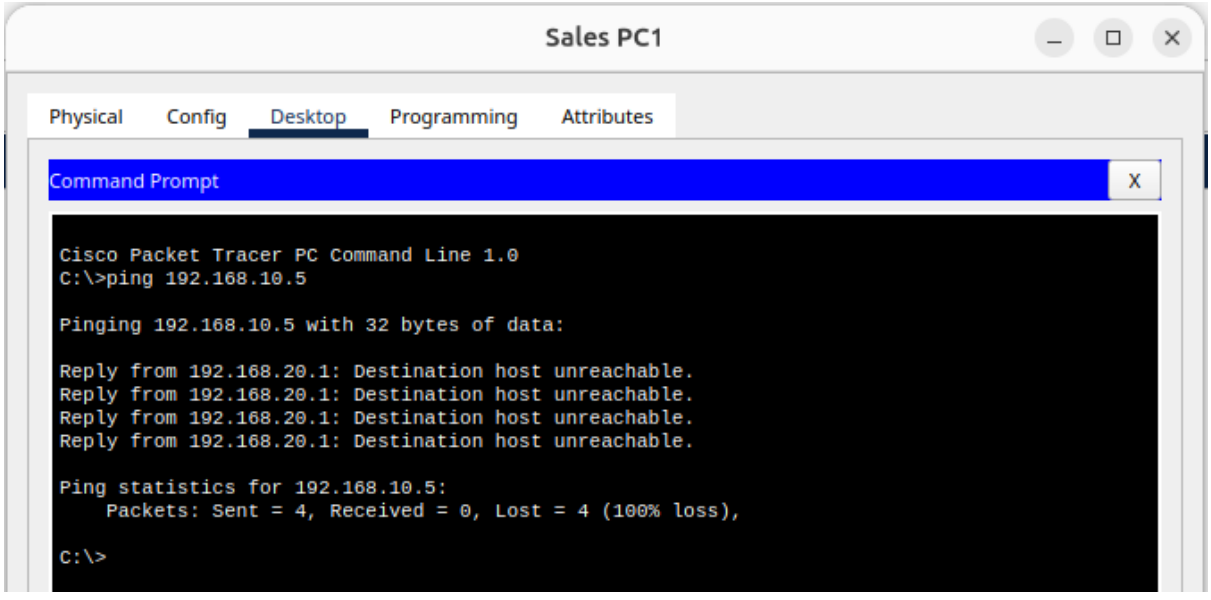
The screenshots above show the IP configuration details obtained from the PCs in the IT, Sales, and HR departments. The configurations confirm that the PCs successfully received their IP addresses, subnet masks, default gateways, and DNS server information from the DHCP server. Each PC was assigned an IP configuration appropriate to its respective departmental network.

Successful Ping from a HR PC1 to IT PC:



The screenshot above shows a successful ping test from one of the PCs in the HR department to a PC in the IT department. Four ICMP packets were sent and all four were successfully received, resulting in 0% packet loss. This confirms that communication between the HR and IT departmental networks is functioning correctly through the router.

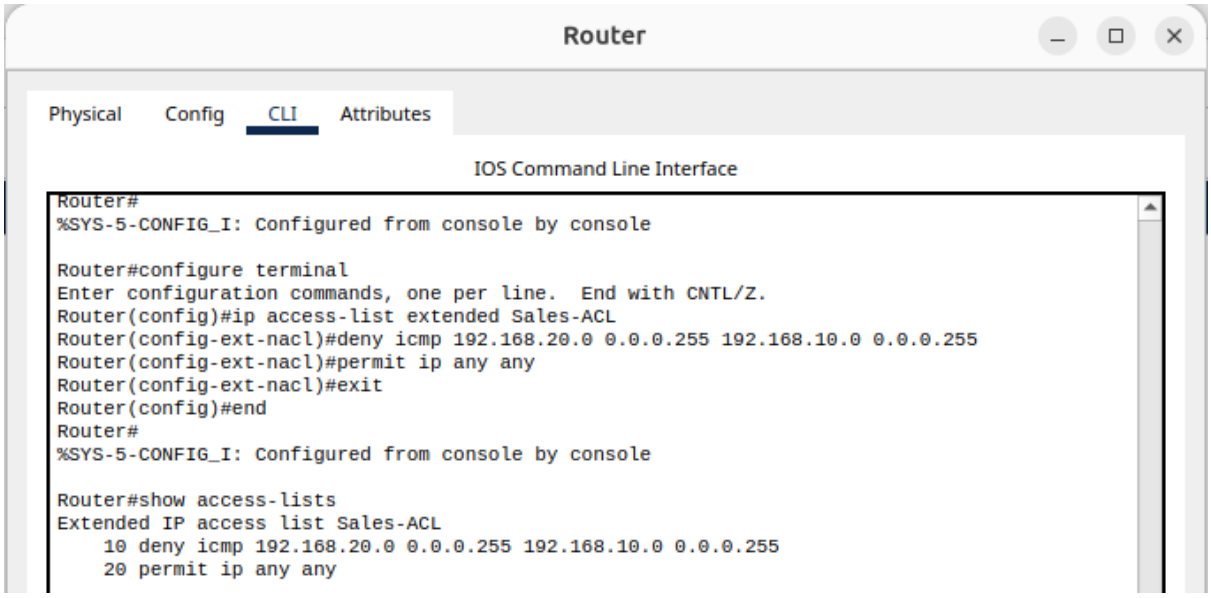
Failed ping from Sales PC to IT PC:



The screenshot above shows a failed ping test from one of the PCs in the Sales department to a PC in the IT department. All four ICMP packets sent were unsuccessful, resulting in 100% packet loss and a “Destination Host Unreachable” message.

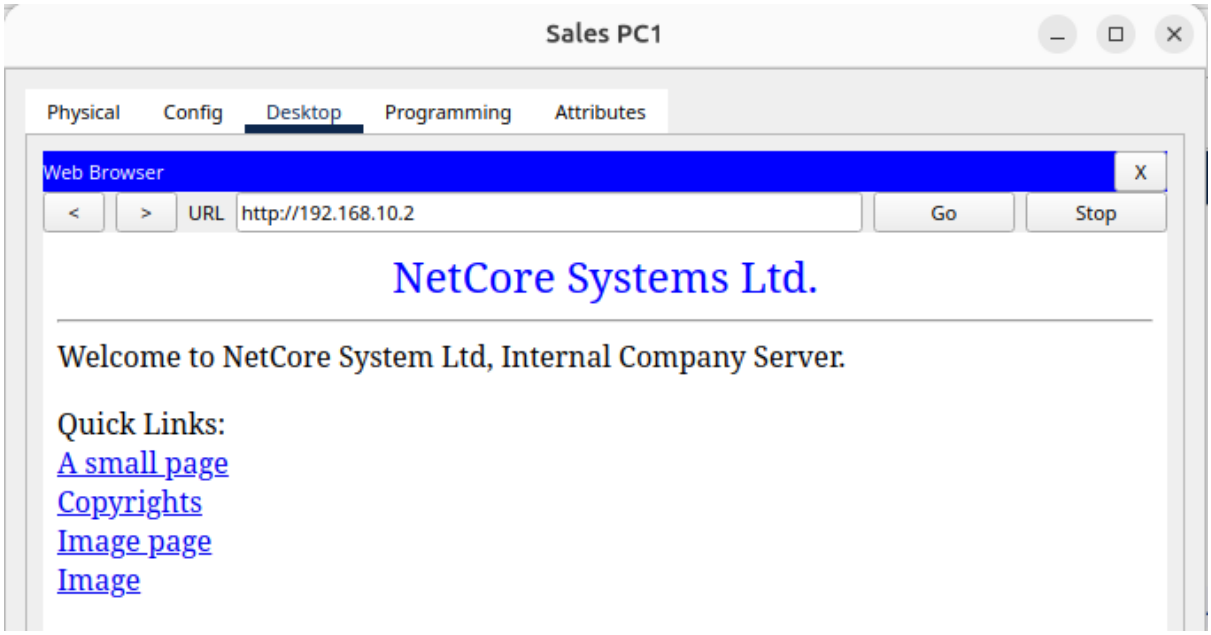
This result is expected and confirms that the network access control policy implemented by TechNova Limited is functioning as intended. Management has restricted ICMP (ping) communication from the Sales network to the IT network as a security measure, while allowing other authorised network services between the two departments.

ACL Command:



The screenshot above shows the Access Control List (ACL) commands entered during the configuration of the router through the Command-Line Interface (CLI). The ACL was configured to restrict ICMP (ping) traffic from the Sales network to the IT network, in accordance with TechNova Limited’s network security policy, while allowing other authorised network services between the two departments.

Sales Department have access to the HTTP service of the Server:



The screenshot above shows that PCs in the Sales department can successfully access the HTTP service provided by the IT server. This confirms that HTTP traffic is permitted between the Sales and IT networks, in accordance with TechNova Limited's network security policy, while ICMP (ping) traffic from Sales to IT remains restricted.